



Software Engineering



Lecture One



2026

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Najwan abed



Software Engineering syllabus

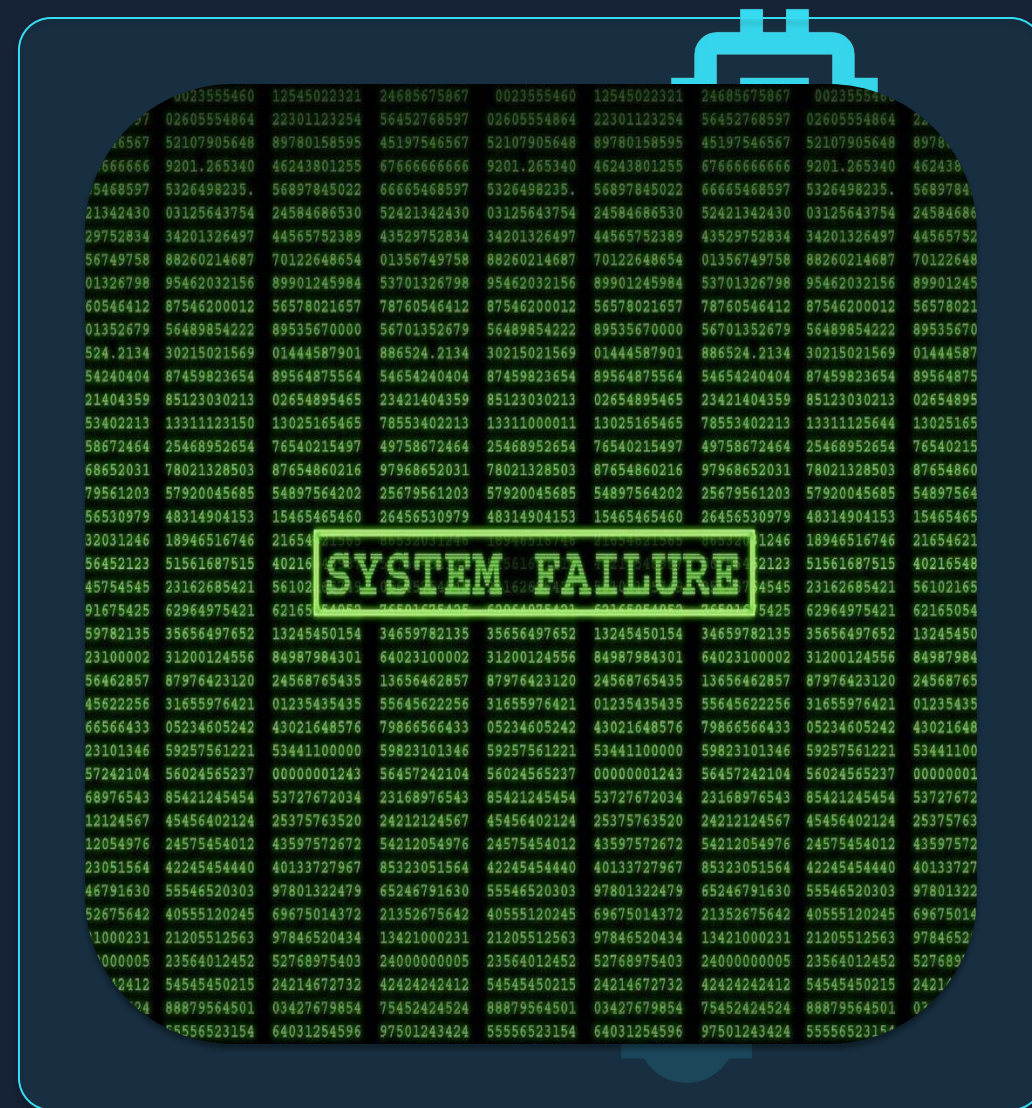
- The 'software Crisis'
- What is software
- Software Characteristics
- Software Applications
- Definitions of software engineering
- Software engineering layers
- The software process
- Software process models
- The linear sequential model
- The prototyping model
- The rad model
- Evolutionary software process models
- The management spectrum
- Testing
- Control graph



The Software Crisis

Software lags far behind
the hardware's potential!

Many large projects fail in that they're late, over budget, don't work well, or are abandoned



How Crisis Manifested



Over-budget

Projects running over-budget



Over-time

Projects running over-time



Inefficient

Software was very inefficient



Low Quality

Software was of low quality



Unmet Requirements

Software often did not meet requirements



Unmanageable

Projects were unmanageable and code difficult to maintain



Famous Software Failures



Therac 25

Radiation therapy machine failure

✖ Cost lives



Arianne 5

Self-destructed 37 seconds after launch

💰 \$370 million losses



Y2K Bug

Millennium date rollover issue

💰 Expensive scare



Pentium FDIV Bug

Floating point division error

\$ Expensive scare



London Ambulance Service (LAS) 1992

Thoroughly documented project failure



London Ambulance Service 1992



2,000-3,000

Calls per day



~66%

Emergency (999) calls



200

Staff (of 2,700 total)



Paper Tickets

Details written on paper tickets for each call



Street Atlas

Map reference lookup in street atlas



Conveyor Belt

Tickets passed to central point



Central Point

Controller de-duplicates tickets



Average Response Time

From call to dispatch

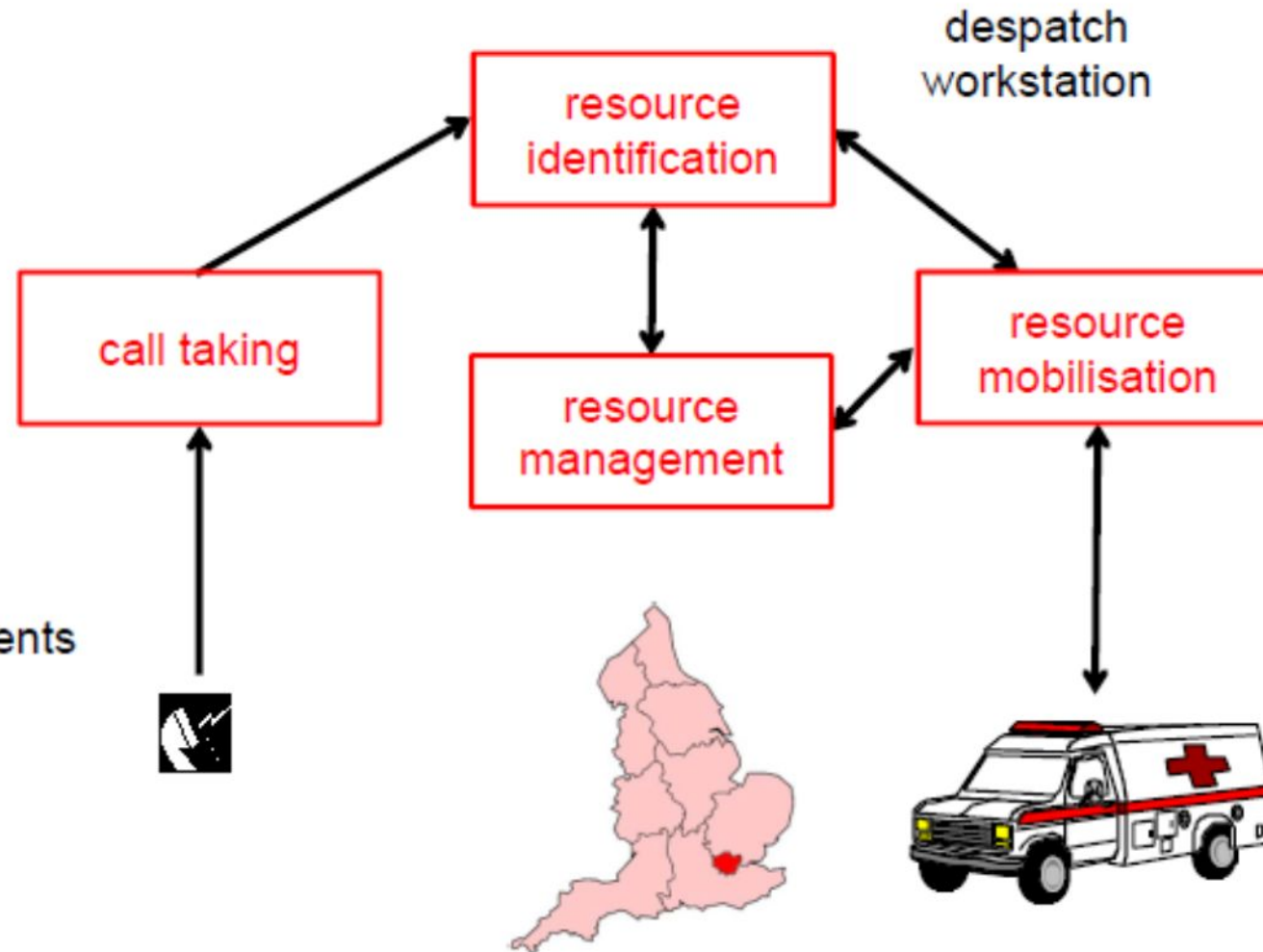
3 min



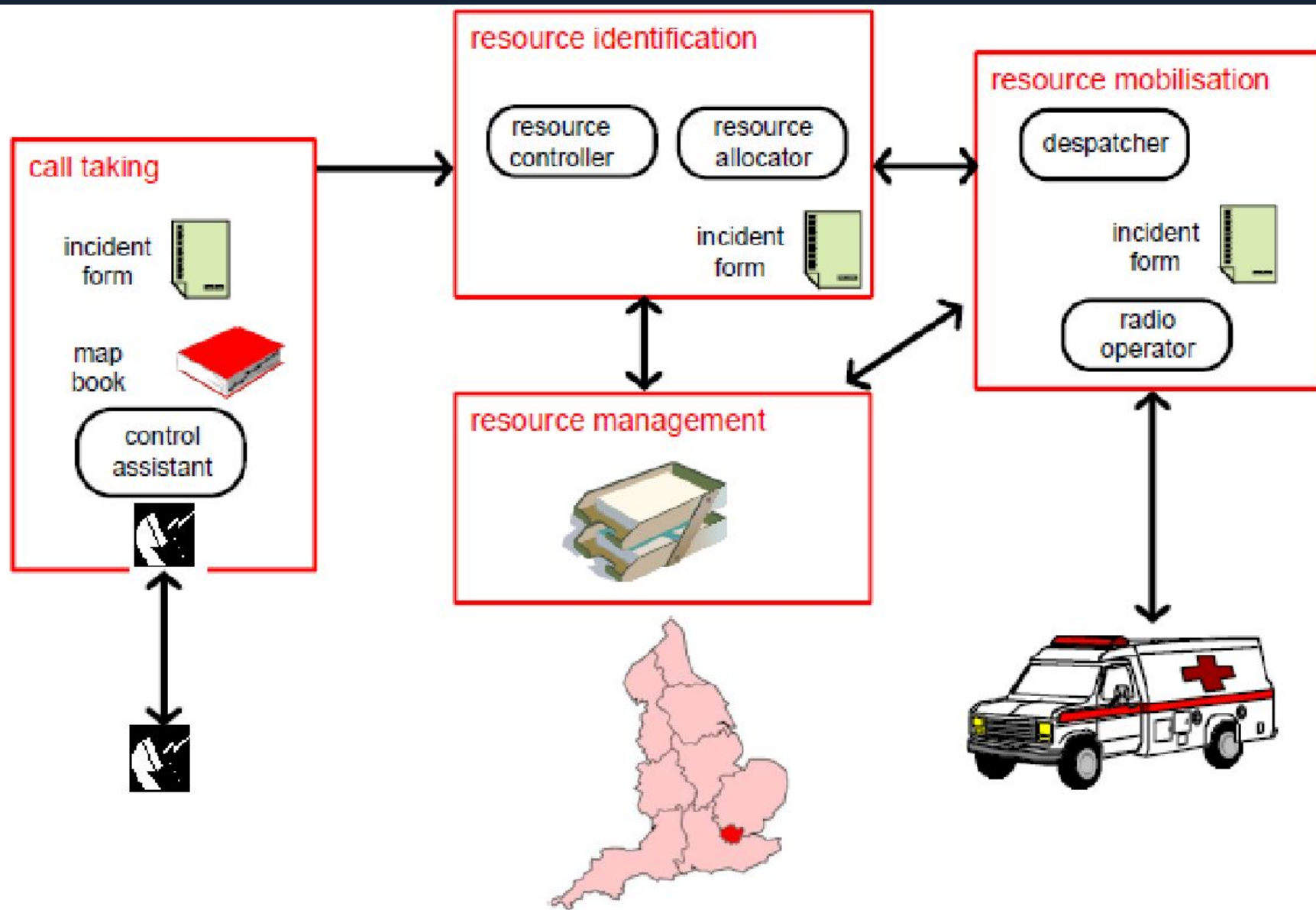
The manual dispatch system

PROPERTIES:

- Large
- Real-time
- Critical
- Data rich
- Embedded
- Distributed
- Mobile components



The manual implementation



Automation Attempt



1980s Failure

Automation attempt failed load test



Awarded To

Systems Options Ltd

Consortium with Apricot & Datatrak



Timeline Issue

Most firms said timeline was unrealistic



Tender Process

→ Feb 1991: Tender sent

→ Jan 1992: Deadline



Market Interest

35

Firms Looked

19

Proposed



Contract Price

£937K

£700K cheaper than next bidder



Project Failures



Ignored Advice

On cost and timescale



No Systems View

Lack of holistic approach



Inflexible Spec

Incomplete specification



Ignored Practices

Established work practices



Unrealistic Timeline

Timescale was unrealistic



Technical Fix

Change through technical system



"They made virtually every mistake in the book"



Paul Williams

LAS Inquiry Member

Common mistakes in software project failures



Understanding Software



Software Characteristics



Intangible

Not physical, exists in digital form



Logic-Based

Comprises algorithms and data structures



Developed Not Manufactured

Created by engineers, not mass-produced



Software Applications



System Software

Operating systems, utilities



Application Software

Business, productivity tools



Engineering Software

CAD/CAM, simulation tools



Embedded Software

Real-time control systems



AI Software

Machine learning, neural networks



Web-Based Software

Cloud services, web apps



Software Engineering

Software Engineering Layers

Definition

The application of engineering principles to software development



Quality Layer

QA focus, testing, verification



Tools Layer

Automated support, CASE tools



Methods Layer

How to do it better, techniques

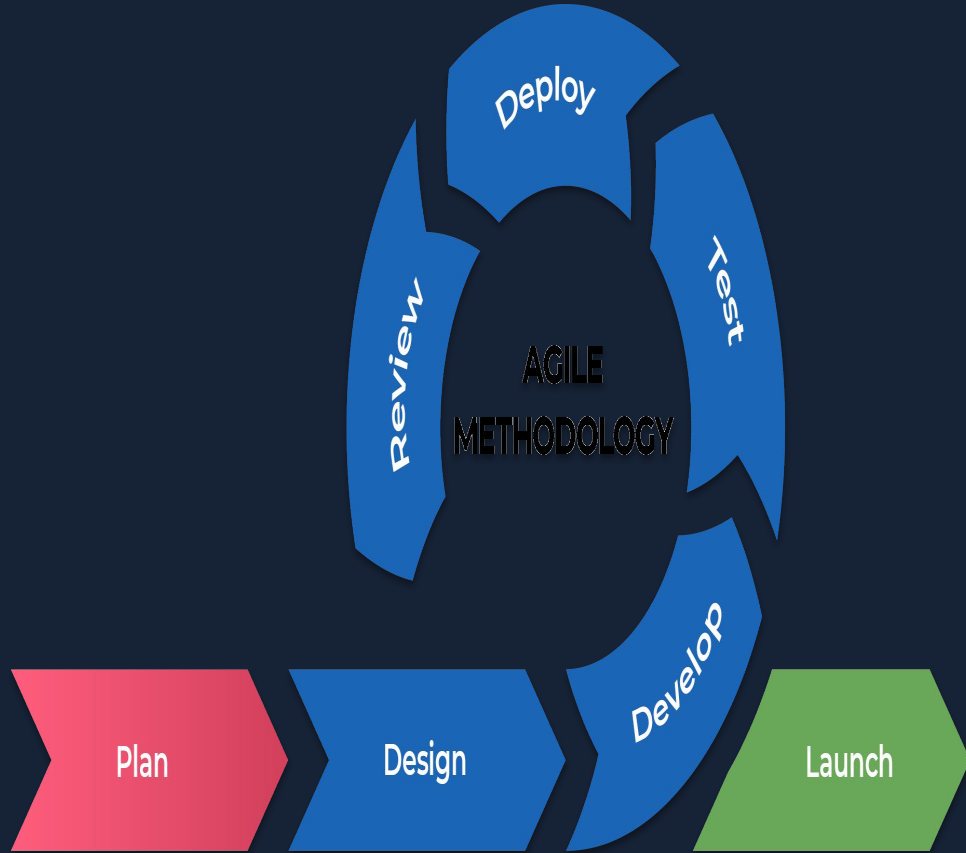


Process Layer

How to do it, workflow models



Software Process Models



Linear Sequential Model

Waterfall approach, sequential phases



Prototyping Model

Rapid iteration, user feedback loop



RAD Model

Rapid Application Development



Evolutionary Models

Incremental, spiral approaches



The Management Spectrum



Process Management

Planning, scheduling, tracking



People Management

Team coordination, motivation



Technology Management

Methods, tools, automation



Quality Management

Testing, QA, standards



Testing

Ensuring software meets quality standards and requirements

- Unit Testing
- Integration Testing
- System Testing
- Acceptance Testing





References & Further Reading



Primary Textbook

Software Engineering Ninth Edition

by Ian Sommerville

Key Topics Covered

Software Crisis

Software Characteristics

Process Models

Testin
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Control Graph

